The complete, self-contained **REVTeX 4.2** manuscript below merges the foundational paper, the four-part Revised Rigorous Architecture, the Bloch–Floquet winding quotient proof, the circuit-QED experimental protocol, and the upgraded \mathbb{C}^N \otimes \mathbb{C}^2 Python simulation in Appendix A. Save the block below as quadrature_time_conjecture.tex and compile directly with pdflatex in Overleaf, TeXShop, or TeX Live.

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%  above for standard two-column Physical Review D layout.)
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\usepackage{mathtools}
\usepackage{amsthm}
\usepackage{hyperref}
\usepackage{listings}
\usepackage{xcolor}

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    linkcolor=blue!70!black,
    citecolor=blue!70!black,
    urlcolor=blue!70!black
}

% Theorem Environments
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\newtheorem{proposition}{Proposition}
\newtheorem{conjecture}{Conjecture}
\theoremstyle{definition}
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\theoremstyle{remark}
\newtheorem{remark}{Remark}

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\begin{document}

%=============================================================================
=========
% TITLE & METADATA
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=========
\title{The Hilbert-Space Quadrature Conjecture:\\
Time as the Wave-Function-Generated Quadrature of Circular Phase}

\author{Steven Michael Allen Owens}
\affiliation{Independent Researcher}

\date{October 18, 2026}

%=============================================================================
=========
% ABSTRACT
%=============================================================================
=========
\begin{abstract}
We formulate and establish the conditional mathematical consistency of
the \emph{Hilbert-Space Quadrature Conjecture}, which proposes that
physical time is not a primitive external background parameter, but an
emergent, state-generated observable arising from the quadratic
mapping (``squaring'') of the intrinsic circular $U(1)$ phase structure
of Hilbert space. Let $H_U = H_C \otimes H_S$ be the bipartite Hilbert space of a
closed quantum universe containing a quantum clock $H_C$ and system $H_S$. To
avoid spectral commensurability singularities, exact relational
```

dynamics is formulated on the universal cover $H_C^{univ} = L^2(R, dt)$ via a Gel'fand triple $\Phi \subset H_U \subset \Phi^\times$ and refined algebraic group averaging $P_{phys} = \delta(\widehat{C})$. We prove that a continuous Bloch--Floquet winding quotient map $\Pi_\omega : L^2(R) \to L^2(S^1)$ pushes the universal cover constraint forward to the compact clock circle S^1, where physical clocks with lower-bounded energy $\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$ reside in the Hardy space $H^2(S^1)$ and resolve the identity via a shift-covariant Holevo phase positive-operator-valued measure (POVM) $E(d\theta)$, thereby circumventing Pauli's theorem. We establish a tri-fold quadrature hierarchy distinguishing the raw clock marginal $M_\Psi(\theta)$, the joint clock-system quadrature $J_\Psi(\theta, F)$, and the conditional system state $\rho_S(\theta)$. In the ideal-clock limit, $\rho_S(\theta)$ satisfies the exact unitary Schr\"odinger equation, while approximate spectral-window wavepackets $|\Psi_w\rangle$ generate peaked semiclassical clock marginals. For realistic clocks with finite energy spread ΔE_C, the conditional quadrature convolution yields an exact Lindblad double-commutator master equation predicting intrinsic energy-basis dephasing at rate $\Gamma_{jk} \sim (E_j - E_k)^2/(2\hbar\Delta E_C)$. Finally, we prove the equivalence of Quadrature Time to Tomita--Takesaki modular flow in the thermal regime, link the framework to Wigner--Weyl phase-space quantization, and propose a circuit-QED transmon-resonator experiment to measure Γ_{jk}.
\end{abstract}

\maketitle

%===
% SECTION I: INTRODUCTION
%===
\section{Introduction: The Problem of External Time and the Phase Fiber}
\label{sec:intro}

In standard non-relativistic quantum mechanics, time $t \in R$ enters the Schr\"odinger equation as a classical background parameter external to the Hilbert space H. As proved by Pauli~\cite{Pauli1933}, any physical system whose Hamiltonian $\widehat{H}$ is semi-bounded below ($\operatorname{Spec}(\widehat{H}) \subset [E_0, \infty)$) admits no self-adjoint time operator $\widehat{T}$ satisfying the canonical Weyl commutation relation $[\widehat{T}, \widehat{H}] = i\hbar I$. Conversely, in canonical quantum gravity and

reparameterization-invariant systems, the Wheeler--DeWitt constraint $\widehat{C}|\Psi\rangle = 0$ eliminates external time completely, leaving a globally stationary universal wave function $|\Psi\rangle$~\cite{DeWitt1967}.

Two foundational programs have sought to recover temporal order from stationary quantum states:
\begin{enumerate}
    \item The \textbf{Page--Wootters mechanism}~\cite{PageWootters1983,Giovannetti2015}, which extracts relational dynamics from quantum entanglement between a clock subsystem H_C and an observed subsystem H_S.
    \item The \textbf{Connes--Rovelli thermal time hypothesis}~\cite{ConnesRovelli1994}, which identifies physical time flow with the state-dependent Tomita--Takesaki modular automorphism group σ_s^{Ψ} of a faithful density state on a von Neumann algebra of observables.
\end{enumerate}

In this paper, we unify and extend these approaches via a geometric and operational framework we call \textbf{State-Generated Quadrature Time} (or the \emph{Hilbert-Space Quadrature Conjecture}). Our starting point is the intrinsic geometry of complex Hilbert space. Normalized pure state vectors form the unit sphere $S(\mathcal{H}) = \{\psi \in \mathcal{H} : \|\psi\| = 1\}$, while physical rays reside in the projective Hilbert space $P(H) = S(H)/U(1)$ ~\cite{Kibble1979,Ashtekar1999}. This quotient defines a principal circular fiber bundle:
\begin{equation}
    U(1) \hookrightarrow S(\mathcal{H}) \xrightarrow{\pi} \mathbb{P}(\mathcal{H}).
    \label{eq:hopf_bundle}
\end{equation}
Along a single ray, the global circular phase $\psi \sim e^{i\theta}\psi$ ($\theta \in S^1 \cong U(1)$) is gauge-redundant. However, when a universal state $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ entangles a clock sector with a system sector, the quadratic Born map---and its phase-space extension, the Wigner--Weyl transform~\cite{Wigner1932}---``squares'' the circular phase amplitudes of $|\Psi\rangle$ into a nonnegative real measure over clock readings and conditional system states. In this picture, \emph{the wave function does not merely evolve within time; the wave function of the clock sector is the physical generator that squares the circular phase geometry of Hilbert space into observable temporal order.}

To render this conjecture mathematically rigorous, our architecture resolves four foundational tensions:
\begin{enumerate}
    \item \textbf{Spectral Singularity Resolution (Sec.~\ref{sec:universal_cover}):} We formulate the exact relational constraint $\widehat{C}|\Psi\rangle = 0$ on the universal cover R via a Gel'fand triple $\Phi \subset H_U \subset \Phi^\times$ and group averaging, and prove that a Bloch--Floquet winding quotient map Π_ω connects R equivariantly to the compact clock circle S^1.
    \item \textbf{Pauli's Theorem Avoidance (Sec.~\ref{sec:povm}):} By restricting the physical clock to the lower-bounded Hardy space $H^2(S^1)$ ($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$), clock readings are represented by a shift-covariant Holevo phase POVM $E(d\theta)$ on S^1 rather than a sharp self-adjoint operator~\cite{Holevo1982,SaleckerWigner1958,PeggBarnett1989}.
    \item \textbf{Tri-Fold Quadrature Separation (Sec.~\ref{sec:quadrature}):} We distinguish the raw clock marginal $M_\Psi(\theta)$ (which is uniform for exact constraint eigenstates and peaked for spectral-window wavepackets $|\Psi_w\rangle$), the joint quadrature $J_\Psi(\theta, F)$, and the conditional system state $\rho_S(\theta)$.
    \item \textbf{Finite-Clock Lindblad Master Equation (Sec.~\ref{sec:lindblad}):} We prove that finite clock energy spread ΔE_C induces a Gaussian phase convolution whose Markovian limit is a completely positive double-commutator master equation with a testable dephasing rate $\Gamma_{jk} \sim (E_j - E_k)^2/(2\hbar\Delta E_C)$.
\end{enumerate}

%==
% SECTION II: UNIVERSAL COVER, RIGGED HILBERT SPACE & WINDING QUOTIENT
%==
\section{Exact Relational Dynamics on the Universal Cover and the Winding Quotient Map}
\label{sec:universal_cover}

\subsection{The Spectral Obstruction on Compact S^1}
Suppose one defines the clock Hilbert space directly on the circle as $H_C = L^2(S^1, d\theta/2\pi)$ with discrete integer spectrum $\widehat{N}|n\rangle = n|n\rangle$ ($n \in Z$) and imposes the exact constraint
\begin{equation}
    \widehat{C}_{\mathrm{naive}}|\Psi\rangle = \left(\hbar\omega \widehat{N} \otimes I_S + I_C \otimes \widehat{H}_S\right)|\Psi\rangle

= 0.
    \label{eq:naive_constraint}
\end{equation}
In the joint eigenbasis $\{|n\rangle_C \otimes |E_k\rangle_S\}$, Eq.~\eqref{eq:naive_constraint} requires $\hbar\omega n + E_k = 0$, which admits nonzero solutions in $H_C \otimes H_S$ if and only if every system energy eigenvalue satisfies the strict commensurability condition $E_k/(\hbar\omega) \in Z$. Equivalently, strict periodicity $|\Psi(2\pi)\rangle = |\Psi(0)\rangle$ forces $e^{-i2\pi\widehat{H}_S/(\hbar\omega)} = I_S$ on the physical subspace. For a generic Hamiltonian $\widehat{H}_S$ with incommensurate energy gaps or continuous spectrum, $\ker(\widehat{C}_{naive}) = \{0\}$ in the strict Hilbert space.

\subsection{Resolution A: Exact Dynamics on the Universal Cover via Group Averaging}
To eliminate this spectral obstruction, we formulate the exact kinematical constraint on the universal cover of the circle,
\begin{equation}
    \mathcal{H}_C^{\mathrm{univ}} = L^2(\mathbb{R}, dt),
\end{equation}
equipped with generalized time eigenstates $|t\rangle$ satisfying $\langle t | t' \rangle = \delta(t - t')$ and self-adjoint clock momentum operator $\widehat{P}_t = -i\hbar\frac{d}{dt}$ with continuous spectrum $\operatorname{Spec}(\widehat{P}_t) = \mathbb{R}$. The universal constraint operator on $H_U^{kin} = H_C^{univ} \otimes H_S$ is
\begin{equation}
    \widehat{C}_{\mathrm{univ}} = \widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_S.
    \label{eq:univ_constraint}
\end{equation}
Because zero lies in the continuous spectrum of $\widehat{C}_{univ}$, physical states are distributional elements of a Gel'fand triple (rigged Hilbert space)~\cite{Marolf1995}:
\begin{equation}
    \Phi \subset \mathcal{H}_U^{\mathrm{kin}} \subset \Phi^\times,
\end{equation}
where $\Phi = S(R) \otimes D(\widehat{H}_S)$ is the dense nuclear space of rapidly decreasing smooth test states. Physical states $|\Psi\rangle \in \Phi^\times$ are constructed via refined algebraic quantization (group averaging) through the rigging map $P_{phys}: \Phi \to \Phi^\times$:
\begin{equation}
    \mathbb{P}_{\mathrm{phys}} = \delta(\widehat{C}_{\mathrm{univ}}) =

\frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} ds \, e^{i s \widehat{C}_{\mathrm{univ}} / \hbar},
    \label{eq:group_averaging}
\end{equation}
which induces the positive-definite physical inner product on the constraint surface:
\begin{equation}
    \langle \phi | \psi \rangle_{\mathrm{phys}} := \int_{-\infty}^{\infty} \frac{ds}{2\pi\hbar} \, \langle \phi | e^{i s \widehat{C}_{\mathrm{univ}} / \hbar} | \psi \rangle_{\mathrm{kin}}.
    \label{eq:phys_inner_product}
\end{equation}
In the generalized clock basis $|t\rangle$, any group-averaged state $|\Psi\rangle = \mathbb{P}_{\mathrm{phys}}|\psi\rangle$ satisfies $\langle t | \widehat{C}_{\mathrm{univ}} | \Psi \rangle = 0$, or
\begin{equation}
    \left(-i\hbar \frac{\partial}{\partial t} + \widehat{H}_S\right) |\Psi(t)\rangle = 0 \implies |\Psi(t)\rangle = e^{-i \widehat{H}_S t / \hbar} |\Psi(0)\rangle,
    \label{eq:univ_evolution}
\end{equation}
for arbitrary system spectra $\operatorname{Spec}(\widehat{H}_S) \subset \mathbb{R}$.

\subsection{The Bloch--Floquet Winding Quotient Map Connecting R to S^1}
To connect the universal cover $H_C^{univ} = L^2(R, dt)$ to an operational cyclic clock on the circle $S^1 \cong \mathbb{R} / (T_0 \mathbb{Z})$ with fundamental period $T_0 = 2\pi/\omega$, we introduce the Bloch--Floquet winding quotient map.

\begin{proposition}[Bloch--Floquet Winding Quotient Homomorphism]
\label{prop:winding}
Let $H_C^{univ} = L^2(R, dt)$ be the universal cover clock Hilbert space with momentum operator $\widehat{P}_t = -i\hbar\frac{d}{dt}$ and generalized basis $\{|t\rangle\}_{t \in \mathbb{R}}$, and let $H_C^{circ} = H^2(S^1) \subset L^2(S^1, d\theta/2\pi)$ be the Hardy space on the circle of period $T_0 = 2\pi/\omega$ with number generator $\widehat{N} = -i\frac{\partial}{\partial\theta}$ ($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$) and clock Hamiltonian $\widehat{H}_C = \hbar\omega\widehat{N}$. Then:
\begin{enumerate}
    \item The periodization map $\Pi_\omega : S(R) \to C^\infty(S^1)$ defined over winding sectors $m \in Z$ by
    \begin{equation}

(\Pi_\omega f)(\theta) := \sum_{m=-\infty}^{\infty} f\!\left(\frac{\theta + 2\pi m}{\omega}\right), \qquad \theta \in [0, 2\pi),
\label{eq:winding_map}
\end{equation}
intertwines temporal translations on R with $U(1)$ rotations on S^1.
\item Under the sign convention $\theta = -\omega t \pmod{2\pi}$ (or conjugated clock phase), $\Pi_\omega \otimes I_S$ maps the universal cover constraint $\widehat{C}_{univ} = \widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_S$ equivariantly onto the circular constraint operator
\begin{equation}
\widehat{C}_{\mathrm{circ}} = \hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S.
\label{eq:circ_constraint}
\end{equation}
\item Summing the universal cover generalized states $|t\rangle$ across all winding sectors $m \in Z$ projected onto the positive-frequency Hardy space $H^2(S^1)$ yields the covariant Holevo phase POVM $E(d\theta) = \frac{1}{2\pi}|\theta)(\theta|d\theta$ on S^1.
\end{enumerate}
\end{proposition}

\begin{proof}
For any Schwartz test function $f \in S(R)$, rapid decay ensures absolute and uniform convergence of the series in Eq.~\eqref{eq:winding_map} on $[0, 2\pi)$, defining a smooth 2π-periodic function on S^1. A shift by $\tau \in R$, $(T_\tau f)(t) = f(t - \tau)$, maps under Π_ω to
\begin{equation}
(\Pi_\omega T_\tau f)(\theta) = \sum_{m=-\infty}^{\infty} f\!\left(\frac{\theta - \omega\tau + 2\pi m}{\omega}\right) = (\Pi_\omega f)(\theta - \omega\tau \pmod{2\pi}),
\end{equation}
proving that Π_ω intertwines the R-action with the $U(1)$-action on S^1. Conversely, given a smooth partition-of-unity window function $w(t) \in C_c^\infty(R)$ supported within a single fundamental domain $(-T_0/2, T_0/2)$ satisfying $\sum_{m\in Z} |w(t - mT_0)|^2 = 1$, the local section lift $(\Pi_\omega^* g)(t) = g(\omega t \pmod{2\pi})\, w(t)$ embeds $L^2(S^1)$ isometrically into a single winding sector of $L^2(R)$.

To establish the push-forward of the constraint operator, differentiate Eq.~\eqref{eq:winding_map} with respect to $\theta = \omega t$ using

the chain rule $\frac{\partial}{\partial\theta} = \frac{1}{\omega}\frac{\partial}{\partial t}$. For a universal cover solution satisfying $-i\hbar \frac{\partial}{\partial t}|\Psi_{\mathrm{univ}}(t)\rangle + \widehat{H}_S |\Psi_{\mathrm{univ}}(t)\rangle = 0$, projecting onto the positive-frequency circular sector where $|\Psi_{\mathrm{circ}}(\theta)\rangle = \sum_{n=0}^\infty e^{-in\theta}|\Psi_n\rangle$ (so that the conjugate phase bra $\langle\theta| = \sum_{n=0}^\infty e^{-in\theta}\langle n|$ extracts positive clock modes $n \geq 0$) yields

\begin{equation}
    i\hbar\omega \frac{\partial}{\partial \theta} |\Psi_{\mathrm{circ}}(\theta)\rangle = \widehat{H}_S |\Psi_{\mathrm{circ}}(\theta)\rangle \iff \left(\hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S\right)|\Psi_{\mathrm{circ}}\rangle = 0.
\end{equation}

Finally, partitioning $t \in R$ into its circular phase $\theta = \omega t \pmod{2\pi}$ and integer winding number $m = \lfloor \omega t / 2\pi \rfloor$, the sum of generalized eigenstates over all winding sectors defines the periodic Dirac comb distribution

\begin{equation}
    |\theta_{\mathrm{comb}}\rangle = \sum_{m=-\infty}^{\infty} \left| t = \frac{\theta + 2\pi m}{\omega} \right\rangle = \sum_{n=-\infty}^{\infty} e^{in\theta} |n\rangle.
\end{equation}

Applying the orthogonal Hardy projection $\mathbb{P}_+ = \sum_{n=0}^\infty |n\rangle\langle n|$ onto physical lower-bounded clock states ($n \geq 0$) maps $|\theta_{\mathrm{comb}}\rangle \mapsto |\theta) = \sum_{n=0}^\infty e^{in\theta}|n\rangle$, wrapping the Lebesgue measure dt into the covariant phase POVM $E(d\theta) = \frac{1}{2\pi}|\theta)(\theta|d\theta$.

\end{proof}

\subsection{Resolution B: Operational Compactifications on S^1}

When working directly on the compact clock space H_C^{circ} without passing to the full universal cover, spectral incommensurability is resolved operationally in one of two ways:

\begin{enumerate}
    \item \textbf{Spectral Window Constraint:} Replace the sharp equality $\widehat{C}|\Psi\rangle = 0$ with a bounded variance condition $\langle \Psi | \widehat{C}^2 | \Psi \rangle \le \epsilon^2$ using a regularized window filter $f(\widehat{C}/\epsilon)$ peaked at zero, yielding normalizable states $|\Psi_\epsilon\rangle = f(\widehat{C}/\epsilon)|\eta\rangle / \|f(\widehat{C}/\epsilon)|\eta\rangle\|$.

    \item \textbf{Finite Pegg--Barnett Clock ($H_C^{(N)} = C^N$):} Truncate the clock Hilbert space to N energy levels $\{|0\rangle, \dots,

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|N-1\rangle\}$ with discrete orthonormal phase
states~\cite{PeggBarnett1989}
    \begin{equation}
        |\theta_k^{(N)}\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1}
e^{in\theta_k} |n\rangle, \qquad \theta_k = \frac{2\pi k}{N}, \quad k
\in \{0, 1, \dots, N-1\}.
        \label{eq:pegg_barnett_states}
    \end{equation}
\end{enumerate}

%=====================================================================
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% SECTION III: COVARIANT CLOCK PHASE POVM
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\section{The Covariant Clock Phase POVM and Avoidance of Pauli's
Theorem}
\label{sec:povm}

Let the physical clock Hilbert space be the Hardy space $H_C = H^2(S^1)$
spanned by orthonormal energy eigenstates $\{|n\rangle\}_{n=0}^\infty$
with lower-bounded Hamiltonian $\widehat{H}_C = \hbar\omega\widehat{N}$
($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$). Following
Holevo~\cite{Holevo1982}, we define the generalized circular phase
vectors $|\theta) \in \Phi_C^\times$ for $\theta \in [0, 2\pi) \cong S^1$:
\begin{equation}
    |\theta) = \sum_{n=0}^{\infty} e^{in\theta} |n\rangle.
    \label{eq:holevo_vectors}
\end{equation}
Because the sum is bounded below by $n = 0$, the states $|\theta)$ are
non-orthogonal. Their distributional inner product is the boundary
Szeg\H{o} kernel on the unit disk:
\begin{equation}
    (\theta | \theta') = \sum_{n=0}^{\infty} e^{in(\theta' - \theta)}
= \pi \delta_{2\pi}(\theta - \theta') + \frac{1}{1 - e^{i(\theta' -
\theta + i0^+)}}.
    \label{eq:szego_kernel}
\end{equation}

\begin{proposition}[Covariant Circular Clock POVM]
\label{prop:povm}
The operator-valued measure $E: B(S^1) \to B(H_C)$ defined on Borel subsets
$\Delta \subset S^1$ by
\begin{equation}
    E(\Delta) = \int_\Delta E(d\theta), \qquad E(d\theta) :=
```

\frac{1}{2\pi} |\theta)(\theta| \, d\theta,
    \label{eq:clock_povm}
\end{equation}
is a normalized, shift-covariant positive-operator-valued measure (POVM) satisfying:
\begin{enumerate}
    \item \textbf{Positivity:} $\langle \chi | E(\Delta) | \chi \rangle = \int_\Delta \frac{d\theta}{2\pi} |(\theta|\chi\rangle|^2 \ge 0$ for all $|\chi\rangle \in \mathcal{H}_C$.
    \item \textbf{Completeness:} $E(S^1) = \int_0^{2\pi} \frac{d\theta}{2\pi} |\theta)(\theta| = I_C$.
    \item \textbf{Phase Shift Covariance:} For the unitary clock generator $U(\phi) = e^{i\phi\widehat{N}}$,
    \begin{equation}
        U(\phi) E(\Delta) U^\dagger(\phi) = E(\Delta \oplus \phi), \qquad \forall \phi \in S^1,
        \label{eq:povm_covariance}
    \end{equation}
    where $\oplus$ denotes addition modulo 2π.
    \item \textbf{Finite-N Weak Convergence:} The finite-dimensional Pegg--Barnett projection-valued measure $E^{(N)}(\Delta) = \sum_{\theta_k \in \Delta} |\theta_k^{(N)}\rangle\langle\theta_k^{(N)}|$ converges in the weak operator topology to $E(\Delta)$ as $N \to \infty$.
\end{enumerate}
\end{proposition}

\begin{proof}
Positivity is immediate from $|(\theta|\chi\rangle|^2 \ge 0$. For completeness, let $|\chi\rangle = \sum_{n=0}^\infty c_n |n\rangle \in \mathcal{H}_C$. By Parseval's theorem for Fourier series in $H^2(S^1)$,
\begin{equation}
    \langle \chi | E(S^1) | \chi \rangle = \int_0^{2\pi} \frac{d\theta}{2\pi} \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} c_n^* c_m e^{i(n-m)\theta} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} c_n^* c_m \delta_{nm} = \sum_{n=0}^{\infty} |c_n|^2 = \langle \chi | \chi \rangle,
\end{equation}
so $E(S^1) = I_C$ by polarization. For phase shift covariance, observe that $U(\phi)|\theta) = \sum_{n=0}^\infty e^{in\phi} e^{in\theta}|n\rangle = |\theta + \phi)$, whence changing variables $\theta' = \theta + \phi \pmod{2\pi}$ in Eq.~\eqref{eq:clock_povm} yields $U(\phi)E(\Delta)U^\dagger(\phi) = E(\Delta \oplus \phi)$. Finally, for any $|\chi\rangle, |\eta\rangle \in \mathcal{H}_C$ and Borel interval $\Delta \subset S^1$, the Riemann sum
\begin{equation}

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    \langle \chi | E^{(N)}(\Delta) | \eta \rangle = \frac{1}{N}
\sum_{\theta_k \in \Delta} \left(\sum_{n=0}^{N-1} c_n^*
e^{-in\theta_k}\right)\left(\sum_{m=0}^{N-1} d_m e^{im\theta_k}\right)
\xrightarrow{N \to \infty} \int_\Delta \frac{d\theta}{2\pi} \langle
\chi | \theta)(\theta | \eta \rangle = \langle \chi | E(\Delta) | \eta
\rangle,
\end{equation}
establishing weak operator convergence.
\end{proof}

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% SECTION IV: QUADRATURE CONJECTURE & CONDITIONAL DYNAMICS
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\section{The Quadrature Conjecture and Tri-Fold Conditional Structure}
\label{sec:quadrature}

To avoid conflating global stationarity with conditional evolution, we
explicitly separate three distinct quadratic objects generated by the
universal state $|\Psi\rangle$.

\begin{definition}[Tri-Fold Quadrature Architecture]
\label{def:trifold}
Let $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ be a state
of the composite universe, let E(dθ) = (1/2π)|θ)(θ|dθ be the covariant clock
POVM on S^1, and let $|\Psi(\theta)\rangle := ((\theta|_C \otimes
I_S)|\Psi\rangle \in \mathcal{H}_S$ be the unnormalized conditional
system vector. We define:
\begin{enumerate}
    \item \textbf{The Raw Clock Marginal ($M_\Psi$):} The probability
density for the clock to register phase θ ∈ S^1 regardless of the system
state:
    \begin{equation}
        M_\Psi(\theta) := \frac{1}{2\pi}
\|\Psi(\theta)\|_{\mathcal{H}_S}^2 = \frac{1}{2\pi} \langle \Psi |
\left(|\theta)(\theta|_C \otimes I_S\right) | \Psi \rangle, \qquad
T_\Psi(d\theta) = M_\Psi(\theta)\,d\theta.
        \label{eq:raw_marginal}
    \end{equation}
    \item \textbf{The Joint Clock-System Quadrature ($J_\Psi$):} For any
system effect or projector F ∈ B(H_S) (0 ≤ F ≤ I_S), the joint probability
density of observing system property F at clock reading θ:
    \begin{equation}
        J_\Psi(\theta, F) := \frac{1}{2\pi} \langle \Psi(\theta) | F |
\Psi(\theta) \rangle = \frac{1}{2\pi} \langle \Psi |
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\left(|\theta)(\theta|_C \otimes F\right) | \Psi \rangle.
        \label{eq:joint_quadrature}
    \end{equation}
    \item \textbf{The Conditional System State ($\rho_S(\theta)$):} The normalized
density operator of the system conditioned on clock phase $\theta$:
    \begin{equation}
        \rho_S(\theta) := \frac{\operatorname{Tr}_C\left[(E(d\theta)
\otimes I_S)|\Psi\rangle\langle\Psi|\right]}{\langle \Psi | E(d\theta)
\otimes I_S | \Psi \rangle} =
\frac{|\Psi(\theta)\rangle\langle\Psi(\theta)|}{\|\Psi(\theta)\|_{\mat
hcal{H}_S}^2},
        \label{eq:cond_state}
    \end{equation}
    such that the conditional probability of outcome $F$ given clock
reading $\theta$ is $P_\Psi(F | \theta) = J_\Psi(\theta, F)/M_\Psi(\theta) =
\operatorname{Tr}_S[\rho_S(\theta) F]$.
\end{enumerate}
\end{definition}

\begin{conjecture}[The Hilbert-Space Quadrature Conjecture]
\label{conj:main}
In a closed quantum universe $H_U = H_C \otimes H_S$, there exists no primitive
external background time parameter. All observable temporal ordering
and dynamical evolution are generated internally by the universal wave
function $|\Psi\rangle$ through the quadratic mapping $\Psi \mapsto
(M_\Psi, J_\Psi, \rho_S)$ over the covariant circular phase POVM $E(d\theta)$.
Classical time $t = \theta/\omega$ is a derived semiclassical statistic of this
state-generated quadrature structure.
\end{conjecture}

\begin{theorem}[Emergence of Conditional Schr\"odinger Dynamics]
\label{thm:schrodinger}
Let $|\Psi\rangle \in \Phi^\times$ be an exact physical state
satisfying the relational constraint
$\widehat{C}_{\mathrm{circ}}|\Psi\rangle = \left(\hbar\omega
\widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S\right)|\Psi\rangle
= 0$ on the positive-spectrum subspace where $E_k = \hbar\omega n_k$ ($n_k \in N_0$), or
equivalently let $|\Psi(t)\rangle$ be the universal cover state of
Eq.~\eqref{eq:univ_evolution} within a single fundamental period
$t = \theta/\omega \in [0, 2\pi/\omega)$. Then:
\begin{enumerate}
    \item The raw clock marginal is globally uniform across the
circle:
    \begin{equation}
        M_\Psi(\theta) = \frac{1}{2\pi} = \mathrm{const.}
        \label{eq:uniform_marginal}
    \end{equation}
```

\item The conditional system state $\rho_S(t) := \rho_S(\omega t)$ is pure, $\rho_S(t) = |\psi(t)\rangle\langle\psi(t)|$, and satisfies the exact unitary Schr\"odinger--von Neumann equation:
\begin{equation}
i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \widehat{H}_S |\psi(t)\rangle, \qquad i\hbar \frac{d}{dt} \rho_S(t) = [\widehat{H}_S, \rho_S(t)].
\label{eq:schrodinger_emergent}
\end{equation}
\end{enumerate}
\end{theorem}

\begin{proof}
Expanding $|\Psi\rangle = \sum_{n=0}^\infty \sum_k c_{n,k} |n\rangle_C \otimes |E_k\rangle_S$ and imposing $\widehat{C}_{\mathrm{circ}}|\Psi\rangle = 0$ yields $(\hbar\omega n - E_k)c_{n,k} = 0$, restricting the support to $n = n_k = E_k/(\hbar\omega)$ with normalized coefficients $\sum_k |\alpha_k|^2 = 1$:
\begin{equation}
|\Psi\rangle = \sum_k \alpha_k |n_k\rangle_C \otimes |E_k\rangle_S.
\label{eq:entangled_stationary}
\end{equation}
Projecting onto the conjugate phase vector $\langle\theta|_C = \sum_{m=0}^\infty e^{-im\theta}\langle m|_C$ gives the conditional amplitude
\begin{equation}
|\Psi(\theta)\rangle = (\langle\theta|_C \otimes I_S)|\Psi\rangle = \sum_k \alpha_k e^{-i n_k \theta} |E_k\rangle_S = \sum_k \alpha_k e^{-i E_k \theta / (\hbar\omega)} |E_k\rangle_S.
\label{eq:psi_theta_expansion}
\end{equation}
Taking the norm squared of Eq.~\eqref{eq:psi_theta_expansion} using the orthonormality $\langle E_j | E_k \rangle_S = \delta_{jk}$ yields
\begin{equation}
M_\Psi(\theta) = \frac{1}{2\pi} \langle \Psi(\theta) | \Psi(\theta) \rangle = \frac{1}{2\pi} \sum_k |\alpha_k|^2 = \frac{1}{2\pi},
\end{equation}
which proves global stationarity of the raw marginal. Setting $t = \theta/\omega$ ($\frac{\partial}{\partial\theta} = \frac{1}{\omega}\frac{\partial}{\partial t}$) and differentiating $|\psi(t)\rangle = |\Psi(\omega t)\rangle$ gives
\begin{equation}
i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = i\hbar\omega \frac{\partial}{\partial \theta} \sum_k \alpha_k e^{-i E_k \theta /

(\hbar\omega)} |E_k\rangle_S = \sum_k \alpha_k E_k e^{-i E_k t / \hbar} |E_k\rangle_S = \widehat{H}_S |\psi(t)\rangle,
\end{equation}
from which $i\hbar\frac{d}{dt}\rho_S(t) = [\widehat{H}_S, \rho_S(t)]$ follows immediately.
\end{proof}

\begin{proposition}[Semiclassical Peaked Time via Spectral-Window Wavepackets]
\label{prop:wavepacket}

Let $w(t) \in S(R)$ be a normalized clock wavepacket ($\int_R |w(t)|^2 dt = 1$) peaked around a semiclassical time t_0 with temporal width ΔT, and define the normalizable history state on $H_C^{univ} \otimes H_S$:
\begin{equation}
    |\Psi_w\rangle = \int_{-\infty}^{\infty} dt \, w(t) \, |t\rangle_C \otimes e^{-i\widehat{H}_S t / \hbar} |\psi_0\rangle_S.
    \label{eq:wavepacket_state}
\end{equation}
Then $|\Psi_w\rangle$ satisfies the spectral-window constraint bound
\begin{equation}
    \|\widehat{C}_{\mathrm{univ}} |\Psi_w\rangle\|_{\mathcal{H}_U} = \hbar \|w'\|_{L^2(\mathbb{R})} \sim \frac{\hbar}{\Delta T},
    \label{eq:window_bound}
\end{equation}
its raw clock marginal is the peaked distribution $M_{\Psi_w}(t) = |w(t)|^2$, and its conditional system state $\rho_S(t)$ still obeys the exact Schr\"odinger equation~\eqref{eq:schrodinger_emergent} everywhere on the support of $w(t)$.
\end{proposition}

\begin{proof}
Applying $\widehat{C}_{univ} = -i\hbar\frac{d}{dt} \otimes I_S + I_C \otimes \widehat{H}_S$ to Eq.~\eqref{eq:wavepacket_state} and integrating by parts gives
\begin{align}
    \widehat{C}_{\mathrm{univ}}|\Psi_w\rangle &= \int_{-\infty}^{\infty} dt \, |t\rangle_C \otimes \left(-i\hbar \frac{d}{dt} + \widehat{H}_S\right)\!\left[w(t) e^{-i\widehat{H}_S t / \hbar} |\psi_0\rangle_S\right] \nonumber\\
    &= -i\hbar \int_{-\infty}^{\infty} dt \, w'(t) \, |t\rangle_C \otimes e^{-i\widehat{H}_S t / \hbar} |\psi_0\rangle_S.
\end{align}
Taking the H_U-norm yields
$\|\widehat{C}_{\mathrm{univ}}|\Psi_w\rangle\|^2 = \hbar^2$

\int_{-\infty}^\infty |w'(t)|^2 dt = \hbar^2 \|w'\|_{L^2}^2$. Projecting onto $\langle t|_C$ gives $|\Psi_w(t)\rangle = w(t) e^{-i\widehat{H}_S t / \hbar}|\psi_0\rangle_S$, so the raw clock marginal is $M_{\Psi_w}(t) = \|\Psi_w(t)\|^2 = |w(t)|^2$, while the normalized conditional state $|\psi(t)\rangle = |\Psi_w(t)\rangle / |w(t)| = e^{-i\widehat{H}_S t/\hbar}|\psi_0\rangle_S$ evolves unitarily under $\widehat{H}_S$.
\end{proof}

%===
% SECTION V: FINITE-CLOCK LINDBLAD DEPHASING
%===
\section{Finite-Clock Corrections and Intrinsic Lindblad Dephasing}
\label{sec:lindblad}

When the clock subsystem is prepared or read out with a finite energy variance $\Delta E_C < \infty$, the Fourier uncertainty relation imposes a fundamental temporal resolution limit:
\begin{equation}
    \sigma_t \gtrsim \frac{\hbar}{2\Delta E_C}.
    \label{eq:clock_uncertainty}
\end{equation}
Consequently, conditioning on an imperfect clock readout at nominal time t averages the ideal unitary evolution $U(t) = e^{-i\widehat{H}_S t/\hbar}$ against the clock pointer's quadrature resolution kernel $G_{\sigma_t}(\tau)$.

\begin{theorem}[Finite-Clock Double-Commutator Master Equation]
\label{thm:lindblad}
Let the observed conditional system density operator under a Gaussian clock resolution kernel $G_{\sigma_t}(\tau) = (2\pi\sigma_t^2)^{-1/2}\exp(-\tau^2/2\sigma_t^2)$ be given by the convolution channel
\begin{equation}
    \rho_S^{\mathrm{obs}}(t) = \int_{-\infty}^{\infty} d\tau \, G_{\sigma_t}(\tau) \, U(t + \tau) \rho_S(0) U^\dagger(t + \tau).
    \label{eq:gaussian_channel}
\end{equation}
Then:
\begin{enumerate}
    \item To second order in σ_t, the short-time conditional state acquires a double-commutator correction:
    \begin{equation}
        \rho_S^{\mathrm{obs}}(t) = U(t)\left[\rho_S(0) -

\frac{\sigma_t^2}{2\hbar^2} [\widehat{H}_S, [\widehat{H}_S, \rho_S(0)]] + \mathcal{O}(\sigma_t^4)\right]U^\dagger(t).
        \label{eq:short_time_expansion}
    \end{equation}
    \item In the coarse-grained Markovian regime where clock phase variance accumulates diffusively as $\sigma_t^2(t) = T_C t$ with characteristic clock correlation scale $T_C \sim \hbar/\Delta E_C$, $\rho_S^{obs}(t)$ satisfies the exact, completely positive Lindblad master equation~\cite{Gambini2004,Milburn1991}:
    \begin{equation}
        \frac{d\rho_S}{dt} = -\frac{i}{\hbar}[\widehat{H}_S, \rho_S] - \frac{T_C}{2\hbar^2} [\widehat{H}_S, [\widehat{H}_S, \rho_S]] = -\frac{i}{\hbar}[\widehat{H}_S, \rho_S] + \frac{T_C}{\hbar^2}\left(\widehat{H}_S \rho_S \widehat{H}_S - \frac{1}{2}\{\widehat{H}_S^2, \rho_S\}\right).
        \label{eq:lindblad_master}
    \end{equation}
    \item In the energy eigenbasis $\widehat{H}_S |E_j\rangle = E_j |E_j\rangle$, diagonal populations $\rho_{jj}(t) = \rho_{jj}(0)$ are conserved, while off-diagonal coherences decay exponentially as
    \begin{equation}
        \rho_{jk}^{\mathrm{obs}}(t) = \rho_{jk}(0) \exp\left(-i\omega_{jk} t - \Gamma_{jk} t\right), \qquad \Gamma_{jk} = \frac{T_C}{2\hbar^2}(E_j - E_k)^2 \sim \frac{(E_j - E_k)^2}{2\hbar \Delta E_C},
        \label{eq:dephasing_rate}
    \end{equation}
    where $\omega_{jk} = (E_j - E_k)/\hbar$.
\end{enumerate}
\end{theorem}

\begin{proof}
Factoring $U(t+\tau) = U(t)U(\tau)$ in Eq.~\eqref{eq:gaussian_channel} and expanding $U(\tau)\rho_S(0)U^\dagger(\tau)$ via the Baker--Campbell--Hausdorff series in τ gives
\begin{equation}
    U(\tau)\rho_S(0)U^\dagger(\tau) = \rho_S(0) + \frac{\tau}{i\hbar}[\widehat{H}_S, \rho_S(0)] - \frac{\tau^2}{2\hbar^2}[\widehat{H}_S, [\widehat{H}_S, \rho_S(0)]] + \mathcal{O}(\tau^3).
\end{equation}

Integrating against $G_{\sigma_t}(\tau)$ using $\int d\tau G_{\sigma_t}(\tau)\tau = 0$ and $\int d\tau G_{\sigma_t}(\tau)\tau^2 = \sigma_t^2$ yields Eq.~\eqref{eq:short_time_expansion}.

To obtain the exact behavior in the energy eigenbasis, expand

$\rho_S(0) = \sum_{j,k} \rho_{jk}(0)|E_j\rangle\langle E_k|$ in Eq.~\eqref{eq:gaussian_channel}:
\begin{equation}
    \rho_{jk}^{\mathrm{obs}}(t) = \rho_{jk}(0) e^{-i\omega_{jk} t} \int_{-\infty}^{\infty} \frac{d\tau}{\sqrt{2\pi}\sigma_t} \exp\left(-\frac{\tau^2}{2\sigma_t^2} - i\omega_{jk}\tau\right) = \rho_{jk}(0) \exp\left(-i\omega_{jk} t - \frac{1}{2}\omega_{jk}^2 \sigma_t^2\right).
    \label{eq:exact_gaussian_fourier}
\end{equation}
Substituting the diffusive clock variance scaling $\sigma_t^2(t) = T_C t$ into Eq.~\eqref{eq:exact_gaussian_fourier} produces Eq.~\eqref{eq:dephasing_rate}. Differentiating Eq.~\eqref{eq:dephasing_rate} with respect to t gives
\begin{equation}
    \frac{d\rho_{jk}^{\mathrm{obs}}}{dt} = \left[-i\omega_{jk} - \frac{T_C}{2}\omega_{jk}^2\right] \rho_{jk}^{\mathrm{obs}}(t).
\end{equation}
Using $[\widehat{H}_S, |E_j\rangle\langle E_k|] = \hbar\omega_{jk}|E_j\rangle\langle E_k|$ and $[\widehat{H}_S, [\widehat{H}_S, |E_j\rangle\langle E_k|]] = \hbar^2\omega_{jk}^2 |E_j\rangle\langle E_k|$, along with the algebraic identity $-\frac{1}{2}[\widehat{H}_S,[\widehat{H}_S,\rho_S]] = \widehat{H}_S\rho_S\widehat{H}_S - \frac{1}{2}\{\widehat{H}_S^2,\rho_S\}$, establishes Eq.~\eqref{eq:lindblad_master} in standard Gorini--Kossakowski--Sudarshan--Lindblad (GKSL) form with a single Hermitian jump operator $L = \sqrt{T_C/\hbar^2}\widehat{H}_S$.
\end{proof}

%===
% SECTION VI: MODULAR FLOW & WIGNER PHASE-SPACE QUADRATURE
%===
\section{Equivalence to Modular Flow and Wigner Phase-Space Quadrature}
\label{sec:modular_wigner}

\subsection{Tomita--Takesaki Modular Flow as State-Generated Quadrature Time}
\begin{proposition}[Modular Time Equivalence]
\label{prop:modular}
Let $|\Psi\rangle = \sum_k \alpha_k |n_k\rangle_C \otimes |E_k\rangle_S$ be a globally stationary bipartite state with non-vanishing Schmidt amplitudes $\alpha_k > 0$. Then the reduced states $\rho_C = \operatorname{Tr}_S(|\Psi\rangle\langle\Psi|)$ and $\rho_S =$

\operatorname{Tr}_C(|\Psi\rangle\langle\Psi|)$ define a Tomita--Takesaki modular operator $\Delta_\Psi = \rho_C \otimes \rho_S^{-1}$ on the support algebra $A_S = B(H_S)$. When the clock-system entanglement spectrum follows a Boltzmann distribution $|\alpha_k|^2 = Z^{-1} e^{-\beta E_k}$, the modular automorphism group $\sigma_s^\Psi(A_S) = \Delta_\Psi^{is}(I_C \otimes A_S)\Delta_\Psi^{-is}$ coincides identically with Heisenberg evolution under Quadrature Time with $t = -\hbar\beta s$.
\end{proposition}

\begin{proof}
Tracing out H_C from Eq.~\eqref{eq:entangled_stationary} gives the faithful reduced system state $\rho_S = \sum_k |\alpha_k|^2 |E_k\rangle\langle E_k|_S$, whose modular Hamiltonian is $K_S = -\ln \rho_S = -\sum_k (\ln|\alpha_k|^2)|E_k\rangle\langle E_k|_S$. Substituting $|\alpha_k|^2 = Z^{-1} e^{-\beta E_k}$ gives $K_S = \beta\widehat{H}_S + (\ln Z) I_S$. Consequently, for any observable $A_S \in A_S$,
\begin{equation}
    \sigma_s^\Psi(A_S) = \rho_S^{is} A_S \rho_S^{-is} = e^{-is K_S} A_S e^{is K_S} = e^{-is\beta \widehat{H}_S} A_S e^{is\beta \widehat{H}_S}.
\end{equation}
Comparing with the conditional Heisenberg evolution $A_S(t) = e^{i\widehat{H}_S t/\hbar} A_S e^{-i\widehat{H}_S t/\hbar}$ induced by Theorem~\ref{thm:schrodinger} establishes exact equivalence under $t = -\hbar\beta s$.
\end{proof}

\subsection{Wigner--Weyl Phase-Space Quadrature (``Squaring the Circle'')}
For a continuous n-dimensional system $H_S = L^2(R^n, d^n q)$, let $\psi(q, \theta) = ((\theta\vert{}_C \otimes \langle q\vert{}_S)\vert{}\Psi\rangle$ be the conditional configuration-space wave function at clock angle $\theta \in S^1$. The conditional Wigner distribution~\cite{Wigner1932} maps the Hilbert-space state quadratically onto the Cartesian phase space $(q,p) \in R^{2n}$:
\begin{equation}
    W_\Psi(q, p; \theta) := \frac{1}{(2\pi\hbar)^n} \int_{\mathbb{R}^n} d^n y \, e^{-i p \cdot y / \hbar} \, \psi\!\left(q + \frac{y}{2}, \theta\right) \psi^*\!\left(q - \frac{y}{2}, \theta\right).
    \label{eq:wigner_quadrature}
\end{equation}
Integrating $W_\psi(q,p;\theta)$ over all phase space $(q,p) \in R^{2n}$ recovers the clock

quadrature density $2\pi M_{\Psi}(\theta) = \int d^n q d^n p W_{\Psi}(q,p;\theta)$, while differentiating with respect to the clock angle θ via Theorem~\ref{thm:schrodinger} yields the relational Moyal equation:

```
\begin{equation}
    \omega \frac{\partial}{\partial \theta} W_\Psi(q, p; \theta) =
\{H_S(q, p), W_\Psi(q, p; \theta)\}_{\mathrm{Moyal}}
\xrightarrow{\hbar \to 0} \{H_S, W_\Psi\}_{\mathrm{Poisson}}.
    \label{eq:moyal_relational}
\end{equation}
```

Thus, $\vert{}\Psi\rangle$ simultaneously squares the circular clock fiber S_C^1 into the temporal measure $M_{\Psi}(\theta)$ and squares the system Hilbert space into the symplectic phase-space arena R^{2n}.

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%=========================================================================
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% SECTION VII: CIRCUIT-QED EXPERIMENTAL PROTOCOL
%=========================================================================
=========
\section{Experimental Protocol in Superconducting Circuit QED}
\label{sec:experiment}
```

Whereas photonic polarization experiments have demonstrated the ideal kinematic Page--Wootters mechanism in two-qubit systems~\cite{Moreva2014}, testing the Hilbert-Space Quadrature Conjecture requires measuring the finite-clock Lindblad dephasing rate Γ_{jk} (Theorem~\ref{thm:lindblad}) as an explicit function of the clock's quantum energy spread ΔE_C. We propose a superconducting circuit-QED realization:

```
\begin{enumerate}
    \item \textbf{Bipartite Hardware ($H_C \otimes H_S$):} Couple a
```

frequency-tunable transmon qubit ($H_S = C^2$, transition frequency $\omega_s/2\pi \approx 5.0 GHz$) dispersively to a high-Q 3D microwave cavity or coplanar waveguide resonator (H_C, frequency $\omega_c/2\pi \approx 6.0 GHz$, photon number operator $\widehat{N} = a^\dagger a$). In the dispersive regime, the interaction Hamiltonian is

```
    \begin{equation}
        \widehat{H}_{\mathrm{int}} = \hbar \chi \, a^\dagger a \,
|1\rangle\langle 1|_S,
        \label{eq:dispersive_H}
    \end{equation}
```

with dispersive shift $\chi/2\pi \approx -1.0 MHz$. Here, the resonator acts as the quantum clock whose phase rotation tracks the relational evolution

of the transmon with effective system energy splitting $E_1 - E_0 = \hbar\chi_{eff}$.

    \item \textbf{Clock Energy Spread Preparation:} Displace the resonator clock into a coherent state $\vert{}\alpha\rangle_C = e^{-\vert{}\alpha\vert{}^2/2}\sum_{n=0}^\infty \frac{\alpha^n}{\sqrt{n!}}\vert{}n\rangle_C$ with tunable mean photon number $\bar{n} = \vert{}\alpha\vert{}^2 \in [10, 10^4]$ and Poissonian number standard deviation $\Delta n = \sqrt{\bar{n}}$, fixing the clock energy spread at

    \begin{equation}
        \Delta E_C = \hbar \omega_c \sqrt{\bar{n}}.
    \end{equation}

    Simultaneously prepare the transmon in the equatorial superposition $\vert{}\psi_0\rangle_S = \frac{1}{\sqrt{2}}(\vert{}0\rangle + \vert{}1\rangle)_S$ via a $\pi/2$ pulse.

    \item \textbf{Quadrature Readout and Scaling Law:} Perform phase-sensitive heterodyne detection on the cavity leakage field (implementing the covariant phase POVM $E(d\theta)$ on S^1) conditioned alongside dispersive state tomography of the transmon's off-diagonal coherence $\vert{}\rho_{01}(\theta)\vert{}$. By Theorem~\ref{thm:lindblad}, the intrinsic clock-induced dephasing rate scales inversely with $\sqrt{\bar{n}}$:

    \begin{equation}
        \Gamma_{01}^{\mathrm{clock}}(\bar{n}) = \frac{(E_1 - E_0)^2}{2\hbar \Delta E_C} \propto \frac{\chi^2}{2\omega_c \sqrt{\bar{n}}}.
        \label{eq:circuit_qed_scaling}
    \end{equation}

    Sweeping $\bar{n}$ across three orders of magnitude while holding cavity temperature and intrinsic qubit T_2^* constant isolates the $\bar{n}^{-1/2}$ quadrature dephasing signature from standard thermal photon shot noise (which increases with $\bar{n}$).
\end{enumerate}

%==========
% SECTION VIII: CONCLUSION
%==========
\section{Conclusion}
\label{sec:conclusion}

By separating the universal cover constraint $\widehat{C}_{\mathrm{univ}}\vert{}\Psi\rangle = 0$ on $L^2(R)$ from the

```
covariant Holevo phase POVM $E(d\theta)$ on the Hardy space $H^2(S^1)$ via the
Bloch--Floquet winding quotient $\Pi_\omega$, the Hilbert-Space Quadrature
Conjecture provides a mathematically self-consistent foundation for
emergent quantum time. Standard unitary Schr\"odinger dynamics emerges
conditionally in the infinite-bandwidth clock limit
(Theorem~\ref{thm:schrodinger}), while physical clocks with finite
energy spread $\Delta E_C$ obey a completely positive double-commutator Lindblad
master equation (Theorem~\ref{thm:lindblad}) that is directly testable
in contemporary circuit-QED architectures.

%====================================================================
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% APPENDIX A: PYTHON NUMERICAL MODEL
%====================================================================
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\appendix
\section{Discrete Numerical Model and Python Implementation ($C^N \otimes C^2$)}
\label{app:python}

To illustrate the convergence of the discrete Pegg--Barnett clock
$H_C^{(N)} = C^N$ to continuous Schr\"odinger evolution as $N \to \infty$, as well as the
exact finite-clock dephasing envelope
$\vert{}\rho_{01}^{\mathrm{finite}}(\theta)\vert{} =
\frac{1}{2}\exp(-\omega_{\mathrm{ratio}}^2 \sigma_\theta^2 / 2)$ where
$\sigma_\theta = (2\Delta n)^{-1}$, the listing below provides a self-contained Python
simulation of the bipartite state $\vert{}\Psi\rangle \in \mathbb{C}^N
\otimes \mathbb{C}^2$.

\begin{lstlisting}[style=pythonstyle, caption={Self-contained Python
simulation of the discrete $C^N \otimes C^2$ Quadrature Time model, demonstrating
exact bipartite conditioning, finite-clock Lindblad dephasing, and
periodic recurrence on $S^1$.}]
import numpy as np
import matplotlib.pyplot as plt

def exact_bipartite_quadrature(N=128, Delta_n=8.0, omega_ratio=1.0):
    """
    Computes the exact conditional quadrature dynamics directly from a
    bipartite entangled state |Psi> in C^N (clock) x C^2 (system)
    using a covariant smeared clock POVM of energy spread Delta_n.
    """
    theta = 2.0 * np.pi * np.arange(N) / N
    n = np.arange(N)
    n0 = N // 2
```

```
    # 1. Clock Readout POVM envelope with energy spread Delta_n
    #    Intrinsic phase uncertainty: sigma_theta = 1 / (2 * Delta_n)
    env_n = np.exp(-(n - n0)**2 / (4.0 * max(Delta_n, 1e-6)**2))
    env_n /= np.linalg.norm(env_n)

    # 2. Ideal conditional qubit history states |psi_ideal(theta_k)>
    #    |psi(theta)> = (|0> + exp(-i * omega_ratio * theta)|1>) /
sqrt(2)
    psi_ideal_theta = np.zeros((N, 2), dtype=complex)
    psi_ideal_theta[:, 0] = 1.0 / np.sqrt(2.0)
    psi_ideal_theta[:, 1] = np.exp(-1j * omega_ratio * theta) /
np.sqrt(2.0)

    # 3. Intrinsic Clock Pointer Quadrature Profile G(theta - theta')
    pointer_amp = np.fft.ifft(env_n) * np.sqrt(N)
    G_profile = np.abs(pointer_amp)**2
    G_profile /= G_profile.sum()

    # 4. Exact conditional density matrix via cyclic convolution on
S^1
    rho_ideal = np.einsum("ki,kj->kij", psi_ideal_theta,
np.conj(psi_ideal_theta))
    rho_finite = np.zeros_like(rho_ideal)
    for k_idx in range(N):
        weights = np.roll(G_profile, k_idx)
        rho_finite[k_idx] = np.einsum("l,lij->ij", weights, rho_ideal)

    # 5. Analytical Gaussian / Lindblad dephasing prediction
    sigma_theta = 1.0 / (2.0 * Delta_n)
    analytical_coh = 0.5 * np.exp(-0.5 * (omega_ratio *
sigma_theta)**2)

    return theta, G_profile, rho_ideal, rho_finite, sigma_theta,
analytical_coh


if __name__ == "__main__":
    N = 128
    Delta_n = 4.0
    omega_ratio = 2.0

    theta, G_prof, rho_id, rho_fin, sig_th, ana_coh =
exact_bipartite_quadrature(
        N=N, Delta_n=Delta_n, omega_ratio=omega_ratio
    )

    fig, axs = plt.subplots(1, 2, figsize=(12, 4.5))
```

```
    # Panel 1: Clock Pointer Quadrature Profile on S^1
    axs[0].plot(theta, np.fft.fftshift(G_prof), "o-", markersize=3,
color="#1f77b4")
    axs[0].set_xlabel(r"Clock Phase Angle $\theta \in [0, 2\pi)$")
    axs[0].set_ylabel(r"Pointer Quadrature $G_{\sigma_\theta}(\theta - \pi)$")
    axs[0].set_title(rf"Clock Quadrature Kernel ($N = N, \Delta n = Delta_n$)")
    axs[0].grid(True, alpha=0.3)

    # Panel 2: Real Part and Magnitude of Qubit Coherence
rho_01(theta)
    axs[1].plot(theta, np.real(rho_id[:, 0, 1]), "--", color="gray",
alpha=0.7,
                label=r"Ideal $Re[\rho_{01}(\theta)]$")
    axs[1].plot(theta, np.real(rho_fin[:, 0, 1]), "-",
color="#d62728", lw=2,
                label=r"Finite-Clock $Re[\rho_{01}^{obs}(\theta)]$")
    axs[1].axhline(ana_coh, color="#2ca02c", ls=":", lw=1.8,
                   label=rf"Analytical Envelope
$\vert{}\rho_{{01}}\vert{} = {ana_coh:.4f}$")
    axs[1].axhline(-ana_coh, color="#2ca02c", ls=":", lw=1.8)
    axs[1].set_xlabel(r"Emergent Clock Phase $\theta = \omega t$")
    axs[1].set_ylabel(r"Conditional Coherence $\rho_{01}(\theta)$")
    axs[1].set_title("Conditional Schr\"odinger Evolution & Clock
Dephasing")
    axs[1].set_ylim(-0.58, 0.58)
    axs[1].legend(loc="upper right", fontsize=8)
    axs[1].grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig("quadrature_time_simulation.pdf", dpi=300)
    plt.show()
\end{lstlisting}

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% REFERENCES
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